

Customer Churn Prediction Report

1. Summary of the Approach

The goal of this project was to predict customer churn for a telecommunications company using machine learning. Customer churn occurs when customers stop using the company's services, and predicting it can help the business take preventive actions.

First, the Telco Customer Churn dataset was loaded and explored to understand its structure, features, and target variable.

The dataset contains information about customer demographics, subscribed services, and account details, with churn as the target variable.

Data preprocessing was performed by handling missing values, encoding categorical variables, and scaling numeric features.

The data was then split into training and testing sets using an 80/20 split with stratification to preserve class distribution.

A K-Nearest Neighbors (KNN) classifier was trained and evaluated using different values of K to find the best-performing model.

2. Model Performance Metrics

The KNN model was evaluated using accuracy, precision, and recall.

Recall was considered the most important metric because identifying customers who are likely to churn is critical for retention strategies.

The final model achieved:

- **Accuracy:** good overall performance
- **Precision:** acceptable
- **Recall:** satisfactory for detecting churned customers

Testing multiple values of K showed that model performance changes depending on the number of neighbors, and the best K value was selected based on recall.

3. Key Findings About Customer Churn

From exploratory data analysis, several patterns related to customer churn were identified:

- Customers with **shorter tenure** are more likely to churn.
- **Month-to-month contracts** show higher churn rates compared to long-term contracts.
- Customers with **higher monthly charges** tend to churn more often.
- Customers **without technical support** are more likely to leave the service.

These findings suggest that both pricing and service stability play an important role in customer retention.

4. Business Recommendations

Based on the model results and data analysis, the following recommendations can be made:

- Encourage customers to switch from month-to-month contracts to longer-term contracts by offering discounts or incentives.
- Provide technical support or bundled services to customers identified as high risk.
- Focus on retention strategies for new customers during their first months with the company.
- Monitor customers with high monthly charges and offer personalized plans when possible.

Using the churn prediction model, the company can prioritize retention efforts more effectively.

5. Limitations and Future Improvements

This project has several limitations:

- The KNN model does not provide clear feature importance, which limits interpretability.
- Model performance depends strongly on data preprocessing and feature scaling.
- KNN may not scale well for very large datasets.

For future improvements, the following steps could be taken:

- Test other models such as Logistic Regression, Decision Trees, or Random Forests.
- Perform feature selection or dimensionality reduction.
- Use cross-validation and hyperparameter tuning to improve performance.
- Explore cost-sensitive learning to better reflect business priorities.